

- Q.30 What is mesh number in screens and explain in brief significance of mesh number.
- Q.31 List different types of agitators used in mixer. Also make a neat and clean sketch of any one.
- Q.32 Explain working of cyclone separator in detail with neat diagram.
- Q.33 Write a note on shell and leaf filters.
- Q.34 Explain in brief gyratory screen with neat and clean diagram
- Q.35 Explain the mechanism of filtration.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 With neat and labelled sketch explain the construction and working of plate and frame filter press.
- Q.37 Explain the construction, working of Nozzle discharge centrifuge with neat and clean diagram.
- Q.38 With neat and labelled sketch explain the construction and working of a fluid energy mill.

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**3rd Sem / Chem, P &P, Chem Engg. (Spl. Paint Tech.),
Chem Engg. (Spl Polymer Engg)**

Subject:- Mechanical Operations

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Diving force used in filter press is _____
- a) Pressure b) Gravity
- c) Both d) None
- Q.2 Which of the following works on the principle of impact?
- a) Gyratory Mill b) Jaw crusher
- c) Ball mill d) Roll crusher
- Q.3 The material is crushed in gyratory crusher by the action of
- a) Impact b) Compression
- c) Attrition d) Cutting
- Q.4 For breaking of hard rock, we use
- a) Crusher b) Ball mill
- c) Tube mill d) Fluid energy mill

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- Q.5 Which of the following involves vibrations?
 a) Hammer mill b) Ball mill
 c) Roll mill d) Grizzly screen
- Q.6 Which of the following is used for very fine crushing?
 a) Jaw crusher b) Hammer mill
 c) Ball mill d) Dodge crusher
- Q.7 The fluid energy mill involves fluid _____.
 a) Pressure b) Temperature
 c) Concentration d) Density
- Q.8 As the size of ball increase, the size of product _____.
 a) Remains same b) Decreases
 c) Increases d) No effect
- Q.9 What is operating mode of size reduction in a roller mill.
 a) Compression b) Attrition
 c) Impact d) Tearing
- Q.10 Sphericity of a special particle is _____.
 a) Zero b) One
 c) Infinite d) None

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Crushers works on the principle of _____
 (Impact/Compression)

- Q.12 Write one example of filtration.
 Q.13 Name any one type of settling.
 Q.14 Name any one material used for making screens.
 Q.15 Define critical speed of ball mill.
 Q.16 Write name of any one filter media.
 Q.17 What is mesh number.
 Q.18 Write name of any grinder.
 Q.19 Write the formula of sphericity.
 Q.20 Define the term size reduction.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Draw a neat labelled diagram and explain of fluid energy mill.
 Q.22 State and explain Rittinger's law and kick's law.
 Q.23 Write a note on Attrition mill.
 Q.24 Write five differences between blake and doge jaw crushers.
 Q.25 Discuss in detail about open and closed circuit grinding.
 Q.26 Define and explain critical speed of ball mill.
 Q.27 Define screen effectiveness and its expression also.
 Q.28 Define specific surface of mixture.
 Q.29 Draw the flow sheet for closed circuit grinding and describe briefly.